

Under Da Lake

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CSCI 5449 | Project 1

## Description

### *Overview, Genre, & Background*

Under Da Lake (UDL) is a 2D metroidvania-like game about collecting sticks. With enough sticks, the player can unlock abilities which allow them to explore the world more. However, exploring the world means going farther and farther from your dam, and thus, increases the risk of drowning!

UDL is a metroidvania-like game. I add “-like” as limited “Sequence Breaking” is allowed (see: Sequence Breaking).

Other than the information provided in the Description heading, there is no additional background information required to complete the game.

### *Basic Controls*

Use the mouse to direct the player character’s movement. To move, press and hold “Left Mouse Button”, “W”, or “Space”.

### *Game Mechanics*

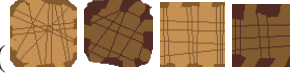
❖ **Movement:** The player can move within the two degrees of freedom allowed by the 2D plane

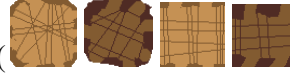
❖ **Pickups:** Sticks, , and the key, , can be picked up and moved by the player.

Bringing a stick to an air area will remove the stick and “score”. Bringing the key to the Door removes the key and “unlocks” the door.



- ❖ **Ice Spikes:** Ice Spikes, , are obstacles to be avoided by the player.



- ❖ **Terrain Destruction:** Hitting a breakable tile ( , scaled in game) with the player's tail breaks the tile and opens a new path (*unlocked via the Breaker upgrade*).



- ❖ **The Door:** The Door, , acts as a guard to the end of the game. Bring the key, , to the door to progress.

- ❖ **Sequence Breaking:** The player can choose to “sequence break”, that is, score a stick from a different screen before the current screen in order (see: Screens as Levels Justification). This is only available for a few sticks in Screens 2 - 4 and involves considerable risk to the player as they must have near perfect movement and navigation in order to avoid drowning.

➤ Author's Note: This mechanic was discovered (or is part of “emergent gameplay”) after much playtesting. I have decided to leave it in the game.

## Rules

- ❖ **“Scoring”:** Bringing a stick to an air area will “score” the stick. Scored sticks count towards upgrades. Sticks can be scored in any order, though level design generally encourages a particular order (see: Sequence Breaking & Screens as Levels Justification)
- ❖ **Drowning & Lung Capacity (LC):** Upon entering water, a red “O2 bar” will appear above the player. The player now has 12 seconds to return to an air area to avoid

drowning. This time is increased by certain upgrades (see: HLC and Breaker upgrade sections).

- ❖ **Pickups:** Picking up a stick will cause the player to lose 35% of their speed. Picking up the key does not affect speed. Pickups cannot be dropped once collected, only replaced by another pickup. This limits the player to one pickup at a time.
- ❖ **Ice Spikes:** Upon hitting one, 10% of the player's remaining LC is removed. This ensures the player cannot directly "die" to an Ice Spike.

- ❖ **Upgrades & The Upgrade Station:** The player must acquire upgrades in order to complete "screens" (see: Screens as Levels Justification). When an upgrade is available, an animation will be played on the Upgrade Station, . Move over the Upgrade Station to unlock the upgrade. Upgrades are locked behind a required score:

- *[5 sticks] Higher Lung Capacity (HLC):* The HLC upgrade applies a 2.5x multiplier to your base LC. This changes LC from 12 sec → 30 sec.
- *[10 sticks] Speed:* The Speed upgrade applies a 1.5x multiplier to your base speed.
- *[15 Sticks] Breaker:* The Breaker upgrade allows the player to destroy certain

textures (, scaled within game) to create new paths. To destroy a breakable tile, the player must hit the tile with their tail. The Breaker upgrade also buffs HLC and Speed by adding 0.2 to the upgrade multipliers. This brings HLC from 30 sec → 32.4 sec, and Speed from 1.5x to 1.7x. The Breaker

upgrade is the last upgrade. Once acquired, the player is able to complete the entire game.

❖ **The Key:** The key spawns within the dam once the player has scored 25 sticks.

### Screens as Levels Justification

As UDL is a metroidvania-like game, it doesn't have levels in the traditional way. Instead the game is split into "screens". For this discussion, they will be labeled as per the diagram below:

|          |          |          |
|----------|----------|----------|
| Screen 1 | Screen 2 | Screen 3 |
| Screen 6 | Screen 5 | Screen 4 |

These screens are labeled in the recommended order of completion. However, this order generally only applies for screens 1-3, more on "sequence breaking" later.

During screens 1 - 3, the player will earn enough sticks to unlock all upgrades. Importantly, the next screen cannot be "cleared" (a screen is "cleared" when all the sticks from it are collected) until the upgrade that screen unlocks is acquired. This does not mean sticks can be completed "out of sequence" (what I will refer to as "sequence breaking"). In fact, some sticks towards the start of each screen can be completed out of sequence; however, the entire screen cannot be cleared.

Screens 1 - 3 can be thought of as the "tutorial levels". The player has to return to the home dam each time they unlock an upgrade, the action of swimming back to a screen with sticks acts as teaching the player the new mechanic. The new mechanic is applied when the

player picks up another stick and needs to make a decision on the best path to return to the dam, which in turn, should take into account the new upgrade in addition to any upgrades previously unlocked.

Screen 4 - 6 can be completed in any order, with the exception of the locked area in Screen 6. The locked area requires the player to gather all 25 “normal” sticks then bring the Key to the Door (see: Pickups & The Door). Upon unlocking the Door, the player must bring the last stick to an air area to complete the game. It is also worth noting that the first stick can be collected without the Breaker upgrade on Screen 4, however, the rest of the screen and following screens cannot be completed without the Breaker upgrade. Screens 4 - 6 act as a “level 2” as all upgrades are required in order to complete the screens.

### **What the game is teaching**

UDL teaches the player to navigate the environment, manage a resource, and understand risks.

Stick locations are placed such that, upon collecting a stick, the other stick(s) is/are visible, or their location is on the next screen in order. This teaches the player to map and plan their route ahead of time while collecting sticks. Upon unlocking the Breaker upgrade, new paths can be unlocked, causing the player to re-adjust their knowledge of the map.

LC is a constant resource the player must manage. Getting lost, having poor movement, or sequence breaking on an un-optimal route will cause the player to drown.

LC ties nicely into understanding how the game teaches risk. Speed, LC, and stick drag (see: Rules > Pickups) all affect how long the player can stay underwater, and thus, which sticks they can collect.

## Changes Between Iterations

Changes in the git repo marked “Playtest” or “Bug Fix” / “BF” (or similar) are the result of playtesting. Below is a table of such changes in chronological order. Changes not marked as such are not the result of playtesting and are thus excluded from this list. Such changes often add (at the time) new untested features, textures, or simply serve to save work for later. Commits have been occasionally edited for spelling and/or grammar. See a full list of changes in the [commits sections of the git repo](#).

| Commit  | Title                                             | Description                                                                                                                        |
|---------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 510260c | PLAYTEST   Changed Game Aspect Ratio              | Changed from 4:3 to 16:9. I think this will be better for level design.                                                            |
| bf2290b | Bug fix   Player doesn't spin out of control      | Player only looks towards mouse when mouse is outside the player sprite                                                            |
| 1e7d3b7 | BF & F   Fixed sprite flipping and added drowning | I now scale the entire sprite object to -1 when I want to flip it.<br><br>Drowning works by starting a timer if you're underwater. |
| 7b76de6 | Playtesting   Removed Sinking                     | After playtesting, I found that the sinking mechanic wasn't fun.<br><br>Also added beaver sprite in this push                      |
| 1ad802e | Playtest   Added Swim Input                       | Added Mouse 1 (left click) as swim input upon playtesting.                                                                         |

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|---------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| b46291a | Playtest   Expanded Dam Area                    | <p>Allowed the player to access the north side of the dam upon playtesting.</p> <p>This reflects the texture better.</p>                                                                                                                                                                                |
| fe2ed72 | Fixed UI bug                                    | Changed UI scene to be a CanvasLayer root instead of a Control root and that fixed the HUD                                                                                                                                                                                                              |
| ddaee4f | Playtest   Fixed Infinite Air Bug               | Upon playtesting it was found that if you touch the air area while in the water, the game would think you're in the air area while being in the water area. I fixed this bug.                                                                                                                           |
| 624d9e8 | Fixed jittery player movement                   | <p>Moved player rotation to _process()</p> <p>Enabled physics interpolation in project settings</p>                                                                                                                                                                                                     |
| fdea6b0 | Playtest   Changed Unlock Order & Increased HLC | <p>On playtesting, I found unlocking speed first is a bit more stressing. I changed it to unlock higher lung capacity first since new players will still be getting used to the controls and game loop.</p> <p>Additionally, I increased the higher lung capacity (HLC) multiplier from 1.5 -&gt; 2</p> |
| fc7cea1 | Playtest   Added screen 1. Increased HLC        | Increased HLC from 2 -> 2.5 to decrease difficulty                                                                                                                                                                                                                                                      |

|         |                                                            |                                                                                                                                                                                                                                                                                                                                                                          |
|---------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9c0b20e | Playtest   Buffed base lung capacity                       | <p>This makes screens all screens less stressful while not overly buffing the HLC upgrade.</p> <p>Also, this push fixes a bug with the HLC upgrade where it would modify the base lung capacity, leading to exponential LC</p>                                                                                                                                           |
| 4a621a4 | Playtest   Nerfed Base LC, Buffed HLC. + Vignette          | <p>Nerfed base LC from 15 sec. -&gt; 12 sec. This was changed in order to slightly nerf the buffed HLC which went from a 2.5x multiplier to 4x. This allows the player to more comfortably explore screen 3 while not being able to beat screen 2 before the HLC upgrade.</p> <p>This commit also added the vignette effect, though it is not fully implemented yet.</p> |
| f9f42a9 | Playtest   Decreased sticks needs for speed upgrade        | Decreased sticks needed from 15 -> 10. This makes screen 3 less of a drag.                                                                                                                                                                                                                                                                                               |
| 0ee4658 | Playtest   Changed US [Upgrade Station] interaction method | Changed US interaction from clicking to simply swimming over it. In playtesting, all players tried to swim/move over the US instead of clicking it.                                                                                                                                                                                                                      |
| 8bfedb0 | Playtest   Nerfed HLC to reflect early speed up            | HLC multiplier from 4x -> 2.5x. A lot of screen 3 can be cleared, but it's impossible to fully clear screen 3 without the speed unlock.                                                                                                                                                                                                                                  |

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| e4f7aed | Playtest   Added mounds to scr1 & 2                | In playtesting, players often chose a "meta" straight line path back to the dam each time. The mounds block this path and add extra length to the optimal path. This also blocks scr3 from being completed before the speed upgrade is acquired. |
| 98da346 | Playtest   Added points until next upgrade UI text | Upon playtesting, I noticed that a lot of players wanted to know when the next upgrade was. Thus, I added this UI text.                                                                                                                          |
| 375a02e | Fixed Ice Spike Penalty                            | Made the lung timer manual to fix the ice spike penalty                                                                                                                                                                                          |
| 8c08f20 | Playtest   Changed breakable sprites               | Added crack lines in the breakable sprites to make it more obvious that they're different from the other tiles.                                                                                                                                  |
| 8be0c00 | Playtest   Buffed HLC and Speed on Breaker upgrade | Also added screen 4.<br><br>Screen 4 was impossible to complete without these buffs. Buffs work by adding 0.2 to all ability multipliers whenever the game detects the breaker upgrade was acquired.                                             |
| 239b25d | Fixed air in giant stick                           | [The player wasn't resetting the LC timer when entering the giant stick's air area. This commit fixes that bug.]                                                                                                                                 |
| 3164323 | Fixed a bug when you drown. Added win screen       | [Points were not being reset when the player drowned. This commit fixes that bug.]                                                                                                                                                               |

|         |                                                                 |                                                                                                                                                                         |
|---------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 273b483 | Bug fix   Fixed ordering bug on lower dam                       | [Fixed a texture ordering issue on the lower dam.]                                                                                                                      |
| 3b81e43 | Bug Fix   Fixed Broken Restart Button                           | The restart button now disables all unlocked upgrades.                                                                                                                  |
| fd50dbb | Bug Fix   Player cannot score BS [Big/Giant Stick] in lower dam | The player was able to score the big stick in the lower dam by leaving and re-entering the air area. This was fixed by adding a StaticBody2D around the lower air area. |